Business Requirements Document (BRD)

Diabetes Readmission Intelligence Platform

Section 1: Executive Summary

Problem Statement: 30-day hospital readmissions for diabetic patients represent a severe clinical and financial burden, costing an average of $15,000 to $17,800 per episode. Under the CMS Hospital Readmissions Reduction Program (HRRP), hospitals face significant penalties, risking up to a 3% reduction in their total Medicare reimbursements for excess readmissions.

Currently, discharge planning decisions rely heavily on subjective clinical intuition or static risk stratification tools (such as the LACE index or HOSPITAL score). These traditional tools lack the necessary accuracy for complex diabetic cases, fail to adapt dynamically to new data, and suffer from the "black box" problem—they may flag a patient as high-risk but fail to provide actionable, patient-specific reasons for that risk.

This project delivers an advanced, dynamic Machine Learning intelligence platform equipped with Explainable AI (XAI) capabilities via SHAP values. By providing real-time, transparent, and personalized readmission risk assessments prior to discharge, the system empowers Discharge Planners to implement targeted, evidence-based interventions. The ultimate goal is to measurably reduce preventable 30-day readmissions, improve patient care transitions, and mitigate CMS financial penalties.

Section 2: Stakeholder Personas

To ensure the platform delivers actionable value, the system is designed around two distinct user personas, each with specific pain points, goals, and operational needs.

Persona 1: Discharge Planner / Care Coordinator (Primary User)

- **Role:** Manages the daily transition of patients from the hospital ward to home or community care settings.
- **Pain Points:** Overwhelmed by high patient volume (e.g., managing 20+ discharges daily).
    - Lacks the time to manually deep-dive into every patient's Electronic Health Record (EHR) to identify subtle readmission risk factors.
    - Traditional risk scores provide a generic number without explaining why the patient is at risk, making it difficult to choose the right intervention.
- **Goals:** Quickly triage today’s discharge list to focus limited time and resources on the highest-risk patients.
    - Understand the specific drivers of a patient's risk (via SHAP explanations) to implement tailored interventions (e.g., medication review, scheduling a 7-day follow-up call, or arranging a community nurse visit).
- **Key Interaction:** Uses the platform daily to review individual patient risk profiles and log clinical interventions.

Persona 2: Hospital Quality / Operations Manager (Secondary User)

- **Role:** Monitors hospital-wide quality metrics, oversees CMS compliance, and allocates budgets for care programs.
- **Pain Points:** * Readmission data is usually retrospective; they learn about readmission spikes and financial penalties weeks or months after they occur.
    - Struggles to pinpoint exactly which clinical departments or diagnosis groups are driving the highest readmission rates.
    - Lacks data to financially justify investments in post-discharge support programs.
- **Goals:** * Monitor the hospital's 30-day readmission rate trend in near real-time against the national benchmark (e.g., 11.2%).
    - Track estimated CMS penalty exposure to drive systemic policy changes.
    - Evaluate the effectiveness of the Discharge Planners' interventions over time.

- **Key Interaction:** Uses the platform weekly or monthly for high-level aggregated reporting, trend analysis, and strategic decision-making.

Section 3: Functional Requirements

The following table outlines the core functionalities the system must deliver to satisfy the needs of the defined personas. Priorities are categorized using the MoSCoW method (Must Have, Should Have, Could Have).

ID	Requirement Description	Priority	Target Persona
FR-01	The system shall generate and display a predictive readmission risk score (0-100) and risk category (Low/Medium/High) for each patient prior to discharge.	Must Have	Discharge Planner
FR-02	The system shall display the top 3 contributing factors (SHAP values) driving the risk score for individual patients in plain, clinical English.	Must Have	Discharge Planner
FR-03	The system shall provide an interactive, sortable list of daily discharges, filterable by age group, diagnosis category, and risk level.	Must Have	Discharge Planner
FR-04	The system shall display an aggregated 30-day readmission rate trend chart compared against the national baseline (11.2%).	Must Have	Manager
FR-05	The system shall visually highlight patients who have not had an HbA1c measurement during their current admission to prompt immediate clinical action.	Should Have	Discharge Planner
FR-06	The system shall calculate and display estimated annual CMS financial penalty exposure based on the current aggregated readmission rate.	Should Have	Manager

FR-07	The system shall include an Agentic AI Chatbot capable of interpreting natural language queries to filter the dashboard data dynamically.	Should Have	Both
FR-08	The AI Chatbot shall utilize a Retrieval-Augmented Generation (RAG) pipeline to answer clinical questions using an embedded knowledge base of scientific research papers.	Could Have	Both

Section 4: Non-Functional Requirements

The system must meet the following technical and performance standards to ensure usability and reliability:

ID	Requirement Description	Category
NFR-01	**Performance:** The web dashboard and its primary visual components must load within 3 seconds under normal network conditions.	Performance
NFR-02	**AI Responsiveness:** The Agentic AI chatbot must process queries and return an answer (including RAG retrieval) within 10 seconds.	Performance
NFR-03	**Compatibility:** The platform must be fully functional and responsive on modern desktop web browsers (Google Chrome, Microsoft Edge, Safari) with a minimum resolution of 1280px.	Usability
NFR-04	**Explainability:** The machine learning architecture must remain transparent; no "black-box" predictions are allowed without corresponding SHAP explanations.	Architecture

Section 5: Out of Scope

To ensure the project remains focused and achievable within the designated timeframe, the following items are explicitly out of scope for this phase:

1. **Live EHR Integration:** The platform will not integrate directly with live Electronic Health Record (EHR) systems (e.g., Epic, Cerner). It will operate on a static, pre-processed dataset (UCI Diabetes).
2. **HIPAA/GDPR Compliance for Live PHI:** Since the system uses heavily de-identified, historical open-source data, complex security protocols for handling live Protected Health Information (PHI) are not implemented.
3. **Native Mobile Application:** Development of dedicated iOS or Android applications is excluded. The focus is on a responsive web-based dashboard.
4. **Dynamic Financial Billing:** The CMS penalty exposure and cost estimates will be calculated using static industry averages (e.g., ~$15,200 per readmission) rather than dynamic, patient-specific billing data.

Section 6: User Stories

The following user stories follow the standard Agile format (`As a [Persona], I want to [Action], so that [Benefit]`) to guide the development of the dashboard features.

US-01: Triage Daily Discharges

- **As a** Discharge Planner,
- **I want to** see today's list of patients ready for discharge sorted by their ML-predicted readmission risk score (0-100),
- **So that** I can prioritize my morning rounds and focus my limited time on the highest-risk patients.
- Acceptance Criteria: Scores must be color-coded (Green < 40, Amber 41-70, Red > 70).

US-02: Understand Risk Drivers (Explainable AI)

- **As a** Discharge Planner,
- **I want to** click on a high-risk patient and see the top 3 specific factors driving their score (e.g., "Patient has 3 prior admissions"),
- **So that** I can choose the most appropriate clinical intervention (e.g., medication review vs. home care).
- Acceptance Criteria: Explanations must use plain English labels mapped from SHAP values, not raw database column names.

US-03: Monitor Quality and Financial Exposure

- **As a** Hospital Quality Manager,
- **I want to** view a trend line of our 30-day readmission rate alongside an estimated CMS penalty exposure,
- **So that** I can report our performance to the board and justify budgets for post-discharge intervention programs.

- Acceptance Criteria: Dashboard must show a benchmark line at 11.2% and calculate estimated financial risk for any rate above the benchmark.

US-04: Natural Language AI Interaction

- **As a** User (Planner or Manager),
- **I want to** ask an AI chatbot questions like "Show me high-risk patients over 65" or "What does research say about HbA1c testing?",
- **So that** I can interact with the data and clinical literature intuitively without needing to learn complex dashboard filters.
- Acceptance Criteria: The chatbot must accurately filter the UI state or retrieve clinical context via the embedded RAG pipeline.

Section 7: Process Flow (As-Is vs. To-Be)

To understand the operational impact of the platform, below is the comparison of the clinical workflow before and after the system implementation.

Current State (As-Is Process)

1. Patient's condition stabilizes and is marked ready for discharge.
2. Doctor writes the standard discharge summary.
3. Discharge Planner reviews the file without any data-driven risk assessment tool.
4. Standard discharge protocol is arranged for **all** patients equally (One-size-fits-all approach).
5. **Result:** High-risk patients go home without enhanced support, leading to preventable readmissions within 30 days and subsequent CMS penalties.

Future State (To-Be Process)

1. Patient is marked ready for discharge.
2. **System Auto-Calculates:** The ML Model automatically calculates the Risk Score & generates SHAP explanations.
3. **Decision Gateway:**
    - **If Risk Score < 70 (Low/Medium):** Patient follows the standard discharge protocol.
    - **If Risk Score >= 70 (High Risk):** System flags the patient on the Discharge Planner’s dashboard.
4. Discharge Planner reviews the patient's specific SHAP risk factors.
5. Discharge Planner selects an evidence-based intervention (e.g., Schedule 7-day follow-up call, Refer to community nurse, or Ensure HbA1c is measured before leaving).
6. **Result:** Targeted resources reduce the 30-day readmission rate, improving patient safety and avoiding financial penalties.

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